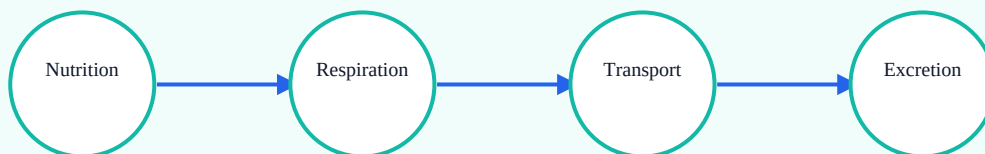


Class 10 Biology

CHAPTER 5 • LIFE PROCESSES

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Life Processes at a Glance



1. What are Life Processes?

Life processes are the basic activities that keep an organism alive and functioning. In Class 10 Biology, the major life processes are **nutrition, respiration, transportation and excretion**.

Why are they needed?

Organisms need energy and raw materials for growth, repair and maintenance. They also need systems to transport useful substances and remove wastes.

Key idea: Food provides materials and energy → respiration releases usable energy → transport distributes substances → excretion removes wastes.

2. Nutrition

Nutrition is the process by which an organism obtains food and uses it for energy, growth and maintenance.

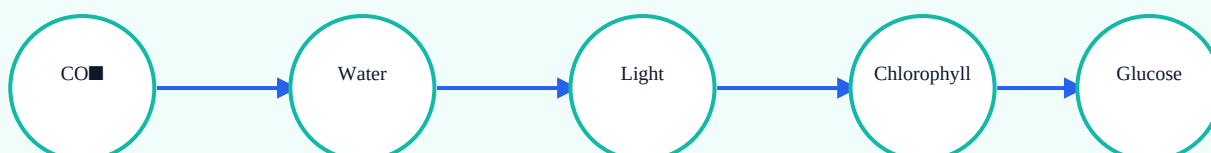
Autotrophic Nutrition

Green plants make their own food by photosynthesis. Chlorophyll traps light energy; carbon dioxide and water are used to form glucose, with oxygen released.

Photosynthesis



Photosynthesis — Main Requirements



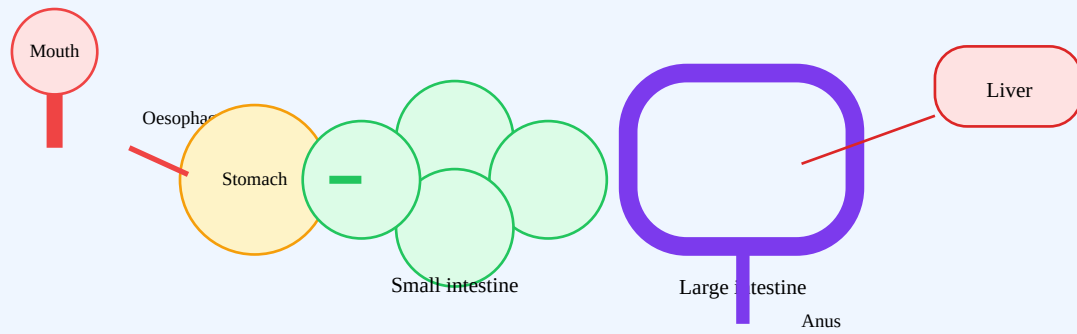
Heterotrophic Nutrition

Organisms that cannot prepare their own food depend on other organisms. Major modes include holozoic, saprophytic and parasitic nutrition.

Human Nutrition

The human digestive system breaks complex food into simpler substances that can be absorbed and used by cells.

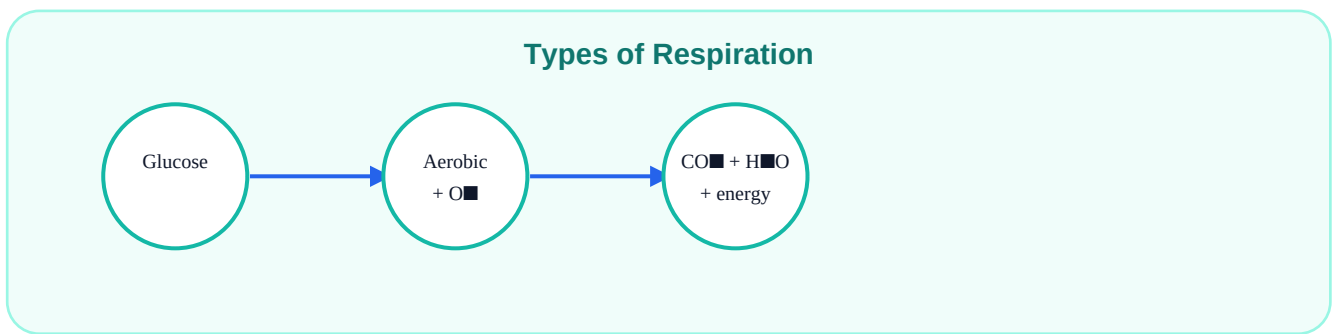
Human Digestive System — Simplified Diagram



Important functions: mouth starts digestion; stomach mixes food and begins protein digestion; small intestine completes digestion and absorbs nutrients; large intestine mainly absorbs water.

3. Respiration

Respiration is the cellular process that releases energy from food. The released energy is used for life activities.



Aerobic respiration

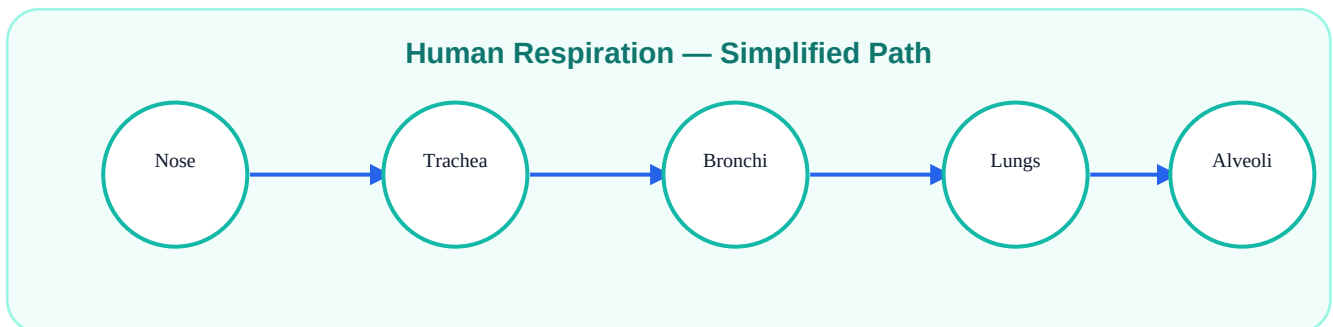
It takes place in the presence of oxygen and generally releases more energy.

Anaerobic respiration

It occurs without oxygen. In yeast, glucose can form ethanol and carbon dioxide; in muscles during oxygen shortage, lactic acid may accumulate.

Gas exchange in humans

Air reaches the lungs through the respiratory passages. In the alveoli, oxygen diffuses into blood and carbon dioxide diffuses from blood into the air spaces.



4. Transportation

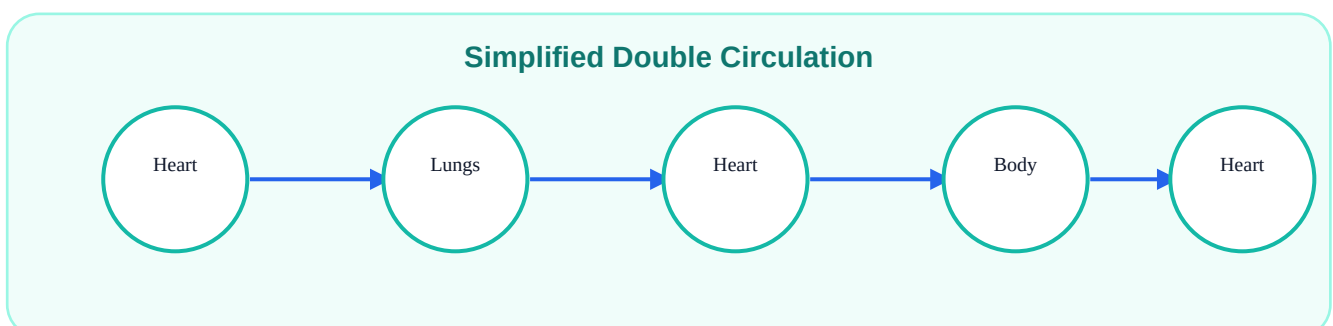
Transportation moves materials such as oxygen, nutrients, hormones and wastes between different parts of the organism.

Transportation in Humans

The circulatory system consists mainly of the **heart, blood and blood vessels**. Arteries generally carry blood away from the heart; veins generally carry blood toward the heart; capillaries allow exchange with tissues.

Double circulation

In humans, blood passes through the heart twice during one complete circulation: pulmonary circulation connects heart and lungs, while systemic circulation connects heart and body tissues.



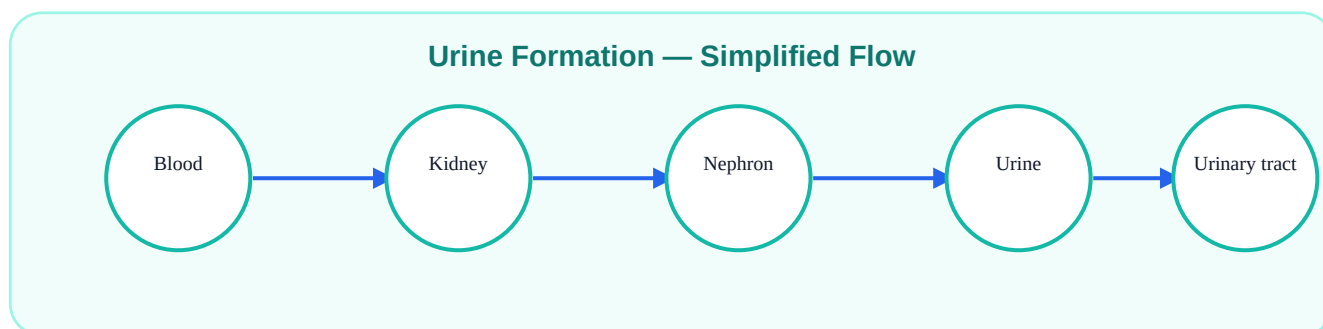
Transportation in Plants

Xylem transports water and minerals mainly from roots toward aerial parts. Phloem transports food from leaves to other parts of the plant.

5. Excretion

Excretion is the removal of metabolic wastes from the body. In humans, the kidneys play a major role in removing nitrogenous wastes and regulating water and salts.

Human Excretory System



Nephron

A nephron is the structural and functional unit of the kidney. It filters blood and helps in selective reabsorption of useful substances and water.

Key terms

- **Filtration:** separation of small molecules from blood into the nephron filtrate.
- **Reabsorption:** useful substances and appropriate amounts of water are taken back into the blood.
- **Urine:** the final fluid containing wastes and excess substances for removal.

6. Quick Revision Table

Process	Main purpose	Key point
Nutrition	Obtain and use food	Photosynthesis in green plants
Respiration	Release energy	Aerobic uses oxygen
Transportation	Move materials	Heart, blood, vessels / xylem, phloem
Excretion	Remove metabolic wastes	Kidneys and nephrons in humans

Exam Focus

1. Define life processes and explain why organisms need them.
2. Write the word/equation for photosynthesis and explain its requirements.
3. Differentiate aerobic and anaerobic respiration.
4. Explain transport in humans and plants.
5. Explain the role of nephron in urine formation.

Note: These are original study notes written from standard Class 10 Biology concepts. They are not a reproduction of any coaching institute's branded PDF.